

Textbook-style Student Notes
BCA30040 – INTRODUCTION TO CLOUD COMPUTING
UNIT–III: CLOUD DEPLOYMENT MODELS

Primary Reference: Thomas Erl, Ricardo Puttini & Zaigham Mahmood, *Cloud Computing: Concepts, Technology & Architecture*

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UNIT–III: CLOUD DEPLOYMENT MODELS : 6

Public Cloud

Private Cloud

Hybrid Cloud

Community Cloud

Choosing the Right Deployment Model

Reference-book basis for Unit-II

Unit-III Topic	Reference Book
Public Cloud	Chapter 4, Section 4.4, p.73
Community Cloud	Chapter 4, Section 4.4, p.74
Private Cloud	Chapter 4, Section 4.4, p.75
Hybrid Cloud	Chapter 4, Section 4.4, p.77
Other deployment models	Chapter 4, Section 4.4, p.78

1. Introduction to Cloud Deployment Models

A **cloud deployment model** describes **where the cloud infrastructure is deployed, who owns or controls it, and who can access it.**

In simple words:

Deployment model tells us who can use the cloud and how the cloud environment is organized.

The four deployment models covered in your syllabus are:

1. **Public Cloud**
2. **Private Cloud**

3. Community Cloud

4. Hybrid Cloud

The reference book places all four under **Chapter 4, Section 4.4 – Cloud Deployment Models**.

Important distinction

Students should not confuse **Service Model** with **Deployment Model**.

Service Model	Deployment Model
Describes what service is provided	Describes how/for whom the cloud is deployed
IaaS	Public
PaaS	Private
SaaS	Community
FaaS	Hybrid

For example:

AWS EC2 + Public Cloud

Here, **EC2** represents the service/infrastructure being consumed, while **Public Cloud** describes the deployment environment.

2. Public Cloud

Meaning

A **Public Cloud** is a cloud environment in which cloud resources are made available to **multiple cloud consumers** through a cloud provider.

The infrastructure is owned and operated by a cloud provider, and customers obtain services/resources according to the provider's offerings.

The reference book defines public clouds as cloud environments made available by third-party cloud providers.

Simple definition

Public Cloud = Cloud infrastructure available to multiple customers through a cloud provider.

Example

Consider **Amazon Web Services (AWS)**.

A company can rent:

- Virtual servers
- Storage
- Databases
- Networking resources

from AWS without purchasing the physical infrastructure.

The same cloud provider can serve many different customers.

Example situation

Suppose **College A** wants to host its website.

Instead of purchasing a server, it uses a public cloud.

College A → AWS → Virtual Server → College Website

The college pays for the resources it uses.

Characteristics of Public Cloud

- Owned/operated by a cloud provider
 - Resources can serve multiple customers
 - Accessible over a network
 - Infrastructure is shared among cloud consumers
 - Resources can be scaled according to requirements
 - Usually uses pay-per-use or subscription-based pricing
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Advantages

1. Lower initial investment

The organization does not need to purchase physical servers.

2. Scalability

Resources can be increased or decreased as required.

3. Easy access

Services can generally be accessed remotely.

4. Provider-managed infrastructure

The provider manages the underlying infrastructure.

Limitation

The organization has less direct control over the physical infrastructure compared with a private cloud.

Remember

Public Cloud = Provider's infrastructure + Many customers

3. Private Cloud

Meaning

A **Private Cloud** is a cloud environment dedicated to a **single organization**.

The infrastructure and resources are used by one organization rather than being made available as a shared environment for unrelated customers.

The organization may operate the private cloud itself or use a provider to operate it.

Simple definition

Private Cloud = Cloud environment dedicated to one organization.

Example

Suppose a bank has strict requirements regarding:

- Customer data
- Security
- Compliance
- Access control

The bank may establish a private cloud for its internal applications.

BANK

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| Private Cloud |

| Servers |
| Storage |
| Applications |
Databases

Only authorized users and systems of the bank access the environment.

Example situation

Imagine MIT-WPU wants to maintain sensitive internal systems such as:

- Student records
- Examination data
- Employee information
- Internal administrative applications

The university could use a private cloud environment for these workloads.

Characteristics

- Dedicated to one organization
 - Greater control
 - Greater control over security policies
 - Can be customized according to organizational requirements
 - Can be located on-premises or managed externally
 - Suitable for sensitive workloads
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Advantages

1. Greater control

The organization can control the cloud environment more closely.

2. Security

Security policies can be customized according to organizational requirements.

3. Customization

Infrastructure can be designed for specific organizational needs.

4. Compliance

Useful where legal or organizational rules require tighter control of data and systems.

Limitations

- Higher cost compared with simply consuming a public cloud service
- Organization may need specialized IT staff
- Infrastructure management can be more complex
- Scaling may require additional planning/investment

Remember

Private Cloud = One organization + Greater control

4. Community Cloud

Meaning

A **Community Cloud** is a cloud environment shared by **multiple organizations that have common requirements or objectives**.

These organizations may have similar:

- Security requirements
- Compliance requirements
- Policies
- Business objectives
- Industry requirements

Simple definition

Community Cloud = Cloud shared by organizations with common requirements.

Example

Consider several hospitals.

All hospitals may have similar requirements for:

- Patient data protection
- Healthcare regulations

- Medical applications
- Data security

They could use a community cloud designed specifically for healthcare organizations.

Hospital A

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Hospital B ---- Community Cloud

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Hospital C

The organizations share the cloud environment because they have common requirements.

Another example

Imagine several government departments need a common cloud environment.

They may have similar requirements regarding:

- Data security
- Government regulations
- Access control
- Data management

A community cloud can be designed around those shared requirements.

Characteristics

- Shared by a specific community of organizations
- Organizations have common concerns
- Common security requirements may exist
- Common compliance requirements may exist
- Resources may be jointly managed or managed by a third party

Advantages

1. Shared cost

Infrastructure costs can be distributed among participating organizations.

2. Common security requirements

The cloud can be designed around the community's security requirements.

3. Common compliance

Useful when organizations follow similar regulations.

4. Collaboration

Organizations can use common infrastructure and services.

Limitation

It may be more expensive and less flexible than a large public cloud because the environment is designed for a specific community.

Remember

Community Cloud = Many organizations + Common requirements

5. Hybrid Cloud

Meaning

A **Hybrid Cloud** combines **two or more different cloud environments**, such as private and public cloud environments, while maintaining their distinct characteristics.

The organization can place different workloads in different environments according to its requirements.

Simple definition

Hybrid Cloud = Combination of different cloud environments working together.

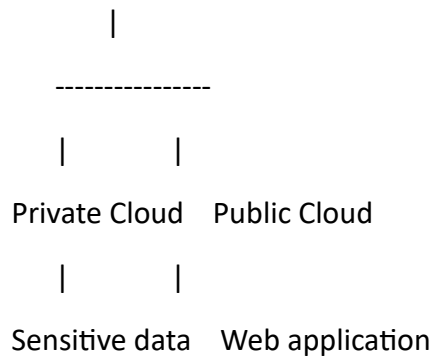
Example

Suppose a bank has:

- Sensitive customer information
- A public-facing banking website

It may keep sensitive information in a **private cloud** while using a **public cloud** for certain scalable web workloads.

BANK



Another simple example

Suppose a college normally uses its private cloud for its student management system.

During examination-result time, thousands of students may access the system simultaneously.

The college can use additional public-cloud resources to handle the increased workload.

This is an example of a **hybrid approach**.

Concept

Normal workload → Private Cloud

Extra workload → Public Cloud

This type of arrangement is often associated with **cloud bursting**, where additional capacity is obtained from another cloud environment when demand increases.

6. Characteristics of Hybrid Cloud

- Combines different cloud environments
 - Workloads can be distributed between environments
 - Can provide greater flexibility
 - Can combine private control with public-cloud scalability
 - Requires appropriate connectivity and integration between environments
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Advantages

1. Flexibility

Different workloads can be placed in different environments.

2. Scalability

Public-cloud resources can provide additional capacity when required.

3. Control

Sensitive workloads can remain in a private environment.

4. Cost optimization

Organizations can choose an appropriate environment for different workloads.

Limitation

Hybrid cloud can be technically complex because the organization must integrate and manage different environments.

Remember

Hybrid Cloud = Combination + Flexibility

7. Comparison of Deployment Models

Feature	Public Cloud	Private Cloud	Community Cloud	Hybrid Cloud
Users	Many customers	One organization	Specific group of organizations	Depends on combined environments
Ownership	Usually cloud provider	One organization/provider	Shared or third party	Multiple environments
Infrastructure	Shared	Dedicated	Shared by community	Combination
Control	Lower	Higher	Medium/High	Depends on components
Cost	Usually lower initial cost	Usually higher	Shared among members	Depends on design
Scalability	Very high	Organization-dependent	Community-dependent	High
Security control	Comparatively lower	Higher	Designed for community	Flexible

Feature	Public Cloud	Private Cloud	Community Cloud	Hybrid Cloud
Best suited for	General workloads	Sensitive workloads	Organizations with common needs	Mixed workloads
Example	AWS public services	Bank's private cloud	Healthcare community cloud	Bank private + public cloud

8. How to Choose the Right Deployment Model?

This is an important part of the syllabus.

The correct deployment model depends on the organization's **requirements**.

Students should ask the following questions.

Question 1: Who needs to use the cloud?

- Many customers → **Public Cloud**
- One organization → **Private Cloud**
- Specific group with common requirements → **Community Cloud**

Question 2: How sensitive is the data?

If the data is highly sensitive, a private environment may be more appropriate.

Question 3: How much control is required?

If the organization needs greater control over infrastructure and security policies, **Private Cloud** may be preferred.

Question 4: Is scalability important?

If very large and changing capacity is required, **Public Cloud** can be useful.

Question 5: Are there different types of workloads?

If some workloads require private control while others require public-cloud scalability:

Hybrid Cloud may be appropriate.

9. Decision-Making Example

Situation

A bank has the following requirements:

1. Customer financial records must remain under strict control.
2. The bank's public website receives unpredictable traffic.
3. The bank needs additional computing capacity during peak periods.

Analysis

Requirement	Suitable choice
Sensitive customer data	Private Cloud
Public website	Public Cloud
Additional capacity	Public Cloud
Combination of both	Hybrid Cloud

Final answer

Hybrid Cloud is suitable because the bank can keep sensitive workloads in its private environment while using public-cloud resources for scalable workloads.

10. Easy Real-Life Analogy

Students often understand deployment models better with a **house analogy**.

Public Cloud

Like staying in a **hotel**.

- Many people use it.
- You don't own the building.
- The hotel manages the infrastructure.

Public = Shared

Private Cloud

Like having **your own house**.

- One family uses it.
- You have greater control.
- You can customize it.

Private = Dedicated

Community Cloud

Like a **housing society**.

- Several families use the society.
- They share common facilities.
- They have common rules.

Community = Shared by a specific group

Hybrid Cloud

Like having **your own house + using a hotel when required**.

- Some needs are handled privately.
- Other needs use shared facilities.

Hybrid = Combination

11. Important Difference: Service Model vs Deployment Model

This is a very common examination question.

Service Model asks:

What am I getting from the cloud?

Examples:

IaaS / PaaS / SaaS

Deployment Model asks:

Who uses the cloud and how is it organized?

Examples:

Public / Private / Community / Hybrid

Example

A company may use:

Public Cloud + IaaS or

Private Cloud + PaaS or

Hybrid Cloud + SaaS

Therefore, **service model and deployment model can be combined**.

12. Memory Pattern for Unit-III

Students can remember the four models using:

Public → Everyone

Private → One organization

Community → Common group

Hybrid → Combination

Public → People/many customers

Private → Personal/one organization

Community → Common requirements

Hybrid → Hybrid/combination

13. Exam-Oriented Summary

Deployment Model	One-line answer
Public Cloud	Cloud environment available to multiple customers through a cloud provider.
Private Cloud	Cloud environment dedicated to a single organization.
Community Cloud	Cloud environment shared by organizations having common requirements.
Hybrid Cloud	Cloud environment combining different cloud environments to meet different requirements.

Most important concept

Deployment model is selected according to requirements such as ownership, access, security, control, scalability, cost and compliance.
